Project: Worked examples
Title: Limits and fits (ISO 286)
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1. Scope and method

Hole and shaft limits for a nominal size and tolerance classes, the resulting fit (clearance, transition or interference) and its extreme values, from the ISO 286-1 formulae.

WHAT IT CALCULATES	RELATION
Standard tolerance factor at the geometric mean D of the size step	i = 0.45·D^(1/3) + 0.001·D (µm)
Width of the tolerance band for the grade	ITn = multiplier × i
Fundamental deviations by letter from the published formulae, and hole deviations by the general rule	EI = −es (A–H); ES = −ei + Δ (K–ZC)
Limits of size of the hole and the shaft	limit = nominal + deviation
Fit extremes, the largest and smallest clearance (negative is interference)	c_{max} = ES − ei; c_{min} = EI − es
Rolling-bearing bore mode	a bearing bore is not an ISO 286 hole; it has its own tolerance (ISO 492) lying below nominal, so ES and EI are entered from the bearing record instead of being computed

Basis: Bases of tolerances, deviations and fits — tolerance grades and fundamental deviations[1](§4, §5); Tables of standard tolerance classes and limit deviations (not reproduced)[2].

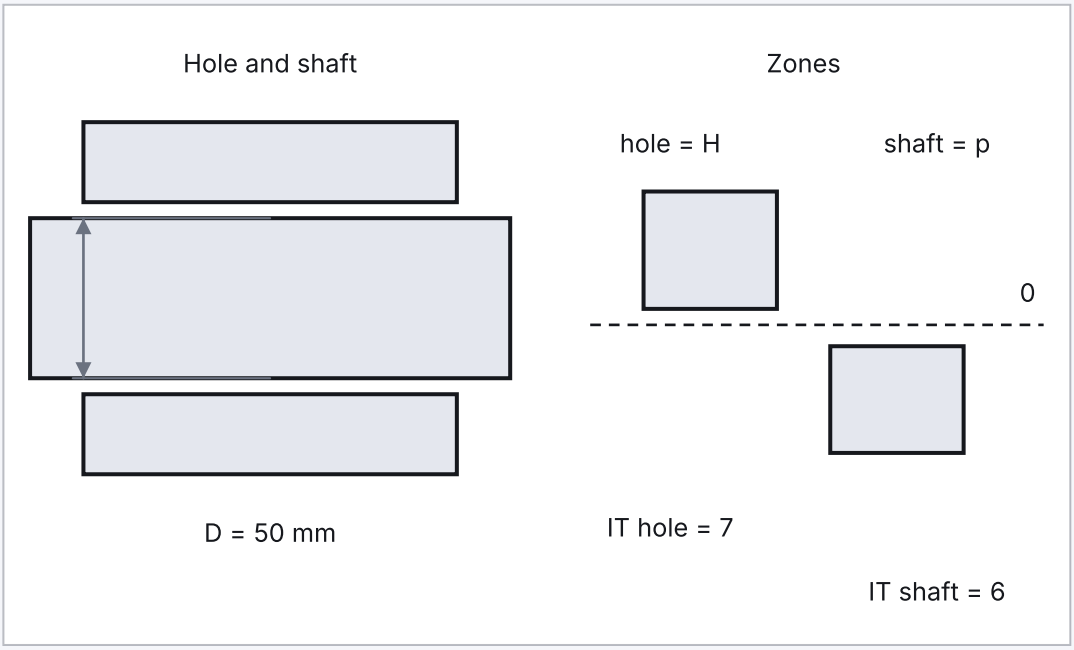


Figure 1 — A hole and its shaft of one nominal size D, and their tolerance zones about the zero line (hole above, shaft below as drawn; the letters place them). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
D	Nominal size	50	mm	drawing: DRG-HUB-04 rev C	1 … 500 mm
$hole_{kind}$	Hole: an ISO 286 class, or a rolling-bearing bore (ES and EI from the bearing record, ISO 492)	ISO 286 hole class		default	any
$hole_{letter}$	Hole fundamental deviation letter	H		spec: DRG-HUB-04: H7/p6	any
$hole_{grade}$	Hole IT grade	7		spec: DRG-HUB-04: H7/p6	5 … 11
$shaft_{letter}$	Shaft fundamental deviation letter	p		spec: DRG-HUB-04: H7/p6	any
$shaft_{grade}$	Shaft IT grade	6		spec: DRG-HUB-04: H7/p6	4 … 11

3. Assumptions

A1. IT values and fundamental deviations are the unrounded results of the ISO 286-1 formulae at the geometric mean of the size step. (The tabulated values apply the standard's own rounding and may differ by about 1 µm for the clearance and transition letters (a to n). For the interference letters p to zc the tables also carry the standard's own increments and sub-steps, so the formula value can be several µm below the tabulated one (p6 at 450 mm: 62.2 against 68 µm), and the sheet warns; enter overrides from a released reference record where the tabulated value governs.)
A2. Hole deviations follow the general rule from the shaft deviations, with Δ = ITn − IT(n−1) where the standard applies it. (Table-specific departures in the standard are not reproduced; use the overrides where they govern.)

4. Working

[7] D_{step} = size step (lower, upper) [1](§4)
= (30, 50] mm
= 38.73 mm
Geometric mean of the size step

[8] i = 0.45·D^(1/3) + 0.001·D [1](§5)
= D = 38.73 mm
= 1.5612 µm
Standard tolerance unit

[9] IT_{hole} = IT7 [1](§5)
= 16 × i
= 24.98 µm
Hole tolerance

[10] IT_{shaft} = IT6 [1](§5)
= 10 × i
= 15.612 µm
Shaft tolerance

[11] es = ei + IT_{shaft} [1](§5)
= 24.98 + 15.612
= 40.592 µm
Shaft upper deviation

[12] ei = fundamental deviation 'p' [1](§5)
= +IT7
= 24.98 µm
Shaft lower deviation
[13] ES = EI + IT_{hole} [1](§5)
= 0 + 24.98
= 24.98 µm
Hole upper deviation
[14] EI = fundamental deviation 'H' [1](§5)
= -es of the same letter
= 0 µm
Hole lower deviation
[15] $D_{hole,max}$ = D + ES
= 50 mm + 24.98 µm
= 50.025 mm
Hole maximum
[16] $D_{hole,min}$ = D + EI
= 50 mm + 0 µm
= 50 mm
Hole minimum
[17] $d_{shaft,max}$ = D + es
= 50 mm + 40.592 µm
= 50.041 mm
Shaft maximum
[18] $d_{shaft,min}$ = D + ei
= 50 mm + 24.98 µm
= 50.025 mm
Shaft minimum
[19] c_{max} = ES - ei
= 24.98 µm − 24.98 µm
= 0 µm
Maximum clearance (negative: minimum interference)
[20] c_{min} = EI - es
= 0 µm − 40.592 µm
= -40.592 µm
Minimum clearance (negative: maximum interference)
[21] i_{max} = es - EI
= 40.592 µm − 0 µm
= 40.592 µm
Maximum interference
[22] i_{min} = ei - ES
= 24.98 µm − 24.98 µm
= 0 µm
Minimum interference (negative: maximum clearance)
[23] fit = fit class
= H7/p6
= "interference"
Fit kind: clearance, transition or interference
[24] c_{mean} = (c_{max} + c_{min}) / 2
= (0 µm + -40.592 µm) / 2
= -20.296 µm
Mean clearance

5. Results

SYMBOL	VALUE	DESCRIPTION
IT_{hole}	25.0 µm	Hole tolerance
IT_{shaft}	15.6 µm	Shaft tolerance
ES	25.0 µm	Hole upper deviation
EI	0 µm	Hole lower deviation
es	40.6 µm	Shaft upper deviation
ei	25.0 µm	Shaft lower deviation
$D_{hole,max}$	50.0 mm	Hole maximum
$D_{hole,min}$	50.0 mm	Hole minimum
$d_{shaft,max}$	50.0 mm	Shaft maximum
$d_{shaft,min}$	50.0 mm	Shaft minimum
fit	"interference"	Fit kind
c_{max}	0 µm	Maximum clearance
c_{min}	-40.6 µm	Minimum clearance
i_{max}	40.6 µm	Maximum interference
i_{min}	0 µm	Minimum interference
c_{mean}	-20.3 µm	Mean clearance

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Fit classified	H7/p6	interference	–	

7. Applicability

WARNING $shaft_{letter}$: the deviation of 'p' is evaluated from the ISO 286-1 formula; the formula value of a letter from p to zc can lie several µm (up to about 9 µm in the 100 to 565 mm checks, p6 at 450 mm 62.2 against 68 µm tabulated) below the tabulated deviation, because the standard's tables add their own increments and sub-steps; the maximum interference is then understated. Enter $ES_{override}$/$EI_{override}$ or $es_{override}$/$ei_{override}$ from a reference record where the table governs. [interference-letter]

Defaults used (inputs left empty):
$hole_{kind}$: not entered, so the default of 'ISO 286 hole class' is used.

8. Verification

```
Template mech.iso286-fit@1.4.1 Content hash 42c6d2c30c0bc9d4 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 1: Basis of Tolerances, Deviations and Fits. ISO 286-1:2010. ISO; 2010.
2. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 2: Tables of Standard Tolerance Classes and Limit Deviations for Holes and Shafts. ISO 286-2:2010. ISO; 2010.